

Job Title: Embedded Systems & Firmware Intern (Industrial Electronics)

Location: Chennai (On-site Preferred): **Internship Duration:** 6–9 Months

Company: Ventura Automation Services

About the Project

Ventura Automation Services is developing next-generation Industrial IoT and AI-based automation products for textile manufacturing. Our engineering team has developed an **industrial-grade embedded controller** based on the **Luckfox Pico Zero (Rockchip RV1106)** platform with a custom-designed PCB for deployment in harsh industrial environments.

The board includes:

- Industrial Ethernet, Wi-Fi Connectivity
- Encoder Interface
- RS-232, RS-485
- RFID Reader Interface, QR/Barcode Scanner Interface (UART)
- EMI/ESD Protection, Industrial Power Supply Design
- Industrial I/O Expansion & Edge AI Capability

The selected interns will work for developing this platform into commercial products used in textile mills and industrial automation applications.

Key Responsibilities

The intern will be involved in:

- Embedded firmware development & Driver development and peripheral integration
- UART, SPI, I²C and GPIO programming
- Ethernet and TCP/IP communication & Wi-Fi configuration and networking
- RS-232 and RS-485 communication protocols
- Encoder interfacing and signal processing
- RFID reader integration & QR/Barcode scanner integration
- Embedded Linux application development & Hardware bring-up and debugging
- Industrial communication troubleshooting
- Bootloader and firmware updates & Performance optimization
- Technical documentation

Required Technical Skills

Candidates should possess strong fundamentals in:

Embedded Programming

- C (Mandatory) & Embedded C (Mandatory)
- Basic C++ & Mirco Python (Preferred)

Microcontrollers & Embedded Systems

Knowledge of one or more:

- Rockchip-based platform
- Arduino, ESP32, STM32
- NXP LPC Series
- Raspberry Pi Pico

Embedded Linux

Basic understanding of:

- Linux Operating System, Shell Commands, Device Tree, UART configuration
- GPIO handling, System services & Cross compilation

Hardware Interfaces

Hands-on exposure to:

- UART, SPI, I²C, GPIO
- ADC, Timers
- Interrupts

Industrial Communication

Basic knowledge of:

- RS-232, RS-485, Modbus RTU
- TCP/IP, Ethernet

Hardware Skills

- Reading schematics, PCB debugging
- Logic analyzer, Signal integrity basics
- Power supply fundamentals

Preferred Skills

Eligibility

Students pursuing final year:

- B.E./B.Tech
- M.E./M.Tech

Preferred disciplines:

- Electronics & Communication Engineering
- Electrical & Electronics Engineering
- Embedded Systems
- Instrumentation Engineering
- Robotics

Desired Candidate Profile

- Have a strong passion for embedded product development.
- Enjoy working with both hardware and firmware.
- Can debug real-world engineering problems methodically.
- Are eager to learn industrial electronics and automation.
- Can work independently while collaborating with a multidisciplinary team.
- Demonstrate curiosity, attention to detail, and a commitment to quality.

Internship Duration

- Minimum: **6 Months**
- Preferred: **9–12 Months**
- Full-time internship preferred.

What You Will Learn

- Industrial embedded product development
- Embedded Linux systems, & Firmware architecture
- Industrial communication protocols & Product testing and reliability engineering
- AI-ready edge computing platforms & Product lifecycle from prototype to production

Internship Deliverables

- Develop production-quality firmware modules & Integrate and validate hardware peripherals.
- Debug communication interfaces and hardware issues & Create reusable drivers and libraries.
- Maintain technical documentation & Demonstrate completed milestones through regular reviews.

Performance Evaluation

- Technical knowledge, Code quality & Hardware debugging skills
- Innovation, Problem-solving ability & Teamwork
- Commitment to project milestones & Documentation

High-performing interns may be offered a **full-time Embedded Firmware Engineer** role upon successful completion of the internship.

Selection Process

1. Resume Screening
2. Embedded C & Electronics Assessment
3. Practical Hardware/Firmware Exercise
4. Technical Interview

Stipend

A **competitive monthly stipend** will be offered based on the candidate's skills, technical assessment, and interview performance.

Why Join Ventura Automation Services?

- Work on a **commercial industrial product**, not just academic prototypes.
- Gain hands-on experience with **custom industrial-grade hardware** designed for factory environments.
- Collaborate directly with Industry Production teams and contribute to products deployed in real manufacturing facilities.
- Build expertise that is highly valued in the embedded systems and industrial automation industries.